GEOMETRY · FOUNDATIONS OF GEOMETRY

Distance & Midpoint Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Distance & Midpoint Explorer** on Mr. K's Classroom: Geometry → Foundations of Geometry → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. For A(1, 2) and B(4, 6), how could a right triangle help you find AB?

   The legs are 3 and 4, so AB is the hypotenuse: √(9 + 16) = 5.

2. Where do you think the midpoint of (–2, 0) and (6, 0) is?

   (2, 0), halfway between –2 and 6.

2 Explore use the tool and record what you see

Distance and midpoint

Drag A and B to each pair of points.

A	B	|Δx|	|Δy|	AB	Midpoint M
(0, 0)	(3, 4)	3	4	5	$(\frac{3}{2}, 2)$
(–4, –3)	(4, 3)	8	6	10	(0, 0)
(–3, 4)	(3, 4)	6	0	6	(0, 4)
(–4, –3)	(1, 2)	5	5	$\sqrt{50} = 5\sqrt{2} \approx 7.07$	$(-\frac{3}{2}, -\frac{1}{2})$
(2, –5)	(–1, –1)	3	4	5	$(\frac{1}{2}, -3)$

Segment addition: A(–4, –3), B(1, 2)

Use the Segment addition tab and drag C.

Where C is	AC	CB	AC + CB
$\frac{1}{2}$ of the way	≈ 3.54	≈ 3.54	≈ 7.07
$\frac{1}{5}$ of the way	≈ 1.41	≈ 5.66	≈ 7.07
$\frac{3}{4}$ of the way	≈ 5.30	≈ 1.77	≈ 7.07

3 Write the rule fill in each blank from what you observed

1. The distance formula comes from the **Pythagorean theorem**: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
2. The coordinates of the midpoint are the **averages** of the endpoints' coordinates.
3. If C is on segment AB, then AC + CB = **AB**.
4. C is the midpoint of AB exactly when AC **=** CB.

4 Check no tool: show your work

1. Find the distance from (–2, 1) to (4, 9).

   √(36 + 64) = 10

2. Find the midpoint of (–2, 1) and (4, 9).

   (1, 5)

3. M(3, –1) is the midpoint of AB and A is (0, 2). Find B.

   (6, –4)

4. C is on AB with AC = x + 4, CB = 2x and AB = 19. Find x.

   3x + 4 = 19, so x = 5

5 Reflect

Why does the distance formula square Δx and Δy?

They are the legs of a right triangle, and the Pythagorean theorem adds the squares of the legs.